

GM EV Battery Architecture Optimization

GAME & ESENG 503 Final Project Report

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Executive Summary

As the electric vehicle (EV) market matures, optimizing the high voltage battery architecture design is crucial to minimizing the overall enterprise cost and maximizing the vehicle program profitability while maintaining customer satisfaction in performance and reliability. Thus, this project paid careful consideration to the cost effectiveness of alternative battery designs over the product life cycle. This study focuses on evaluating two potential battery architectures: a cell to pack design and a cell to module to pack design. It focuses on determining which architecture offers the lowest enterprise cost considering piece price, manufacturing investment, manufacturing scrap, service costs, and recycling.

The approach is to compare an existing cell to pack design against a module based design created by the capstone team, which will be based on benchmarking and advanced GM battery module bill of design concepts. The designs, created using Siemens NX, are to be evaluated for pack energy, mass, piece cost, service cost, manufacturing cost, recycling cost, and system performance.

The evaluation uncovered key trade-offs between the two battery architectures. Pack energy and mass show favor in the cell to pack, and the remaining criteria favor the module based design.

First, the cell to pack reduced bill of material (part count) resulting in a higher pack energy and lower piece price. Second, the cell to pack also offered a reduced manufacturing cost, due to the lower number of assembly steps and stations to produce the battery.

The third criteria, manufacturing scrap, resulted significantly in favor of the module based pack. The module based pack saved cost per pack in mitigated scrap. This is a result of the module pack's ability to limit the scrap costs to a single module. The cell to pack design can result in replacing nearly an entirely completed battery depending on the station the defect occurs in.

Similarly, service cost results in the most favorable business case for the module based design compared to the other criteria. The module based design will require significantly less labor time in service and minimizes the cost of replacement hardware to the parts with a failure. The cell to pack design, however, requires significantly more labor if a failed component is to be replaced. If the failed component is one permanently bonded to the cells (cells, interconnect board and battery tray) this results in a full battery replacement.

Lastly metric six, recycling cost, also favored a module based design for similar reasons to service. The current state of battery recycling technology today limits the size of the parts to be shredded to modules. Thus, for a cell to pack design significant time is required to disassemble the cells from the battery tray.

Combining the life cycle costs results in an overall pack savings for the module based design. As a result the study recommendation, based on the current state of technology, is to pursue a module based design to minimize costs over the program life.

1.0 Introduction and Current State

The General Motors (GM) Advanced & New Product Battery organizations are responsible for developing Rechargeable Energy Storage Systems (RESS) components, subsystems, and battery pack architectures for all future electrified vehicles (EV). In recent years, the organizations have heavily focused on EV profitability. Profitability is maximized by providing the most desirable vehicle to command price while also minimizing total enterprise costs to GM in producing the vehicle. The negotiated RESS targets define customer satisfaction while the design solutions to achieve the targets define the enterprise costs.

The current task is to identify the lowest cost battery architecture for the next generation of mid-size SUVs. Specifically, based on prior studies, to identify if a cell to pack or module based architecture provides the lowest enterprise cost while remaining competitive in the market. Developing the minimum enterprise cost solution to a given set of requirements for EVs is a complex optimization process. The optimization balances component, manufacturing and service costs.

1.1 Overview of the Project

Electric vehicles are a quickly developing and maturing market. Customers have sought out the quiet torque-rich vehicles at an increasing rate. In 2023, there were 3.5 million more global EV sales than in 2022. This is a 35% year-over-year increase, bolstered by government tax credit and a global trend for sustainable purchases.

Currently, 42% of the global electric-SUV market is midsize SUVs. This is the largest segment of electric-SUVs, and therefore is the most desirable to optimize. Creating a more efficient and cost effective battery system will better position GM within the highly competitive EV market.

The capstone project will begin with detailed internal and external benchmarking of these electric midsize SUV battery technologies. Those learnings will be used in conjunction with internal knowledge to design the lowest enterprise cost module based design and compare it against a cell to pack design. The head to head comparison will provide a large set of data to inform future battery program decisions and enable improved future EV profitability.

1.2 Review of Current State Design / Literature Review

As it relates to battery pack construction, there are many ways to theoretically integrate cells into a vehicle. The industry has seen module-based, cell-to-pack, and could be trending towards cell-to-chassis and module-to-chassis architectures in the future. Currently most battery packs are module-based, which means the battery packs are built from groups of cells packed within modules.

The modules are then packed and interconnected into a case that provides thermal management through a metal cooling plate with its own inlet and outlet, physical protection for the cells and an interface with the other modules.

Cell-to-pack batteries seem to be the next generation of pack construction as there are several examples of cell-to-pack batteries currently entering the market. These include the Tesla's 4680 packs, BYD's blade cell to packs, and Qilin's CATL pack. Cell-to-pack removes the modules from the design, and packs cells into the battery structure directly. The main literature based advantages of cell-to-pack are complexity reduction in the bill of materials, manufacturing process step reduction, the increase in volumetric energy efficiency, and cost reduction. In this construction, the sidewalls, tops and bases of modules can be eliminated, freeing up space in the pack and reducing the complexity as highlighted.

Below is an example of a cell-to-pack design utilizing larger prismatic cells. While the image below is from a public website, the team has visited GM benchmarking suppliers, has access to physical properties at GM and access to teardown data from A2Mac1 and Caresoft online portals. These online tools provide the full details on nearly all EVs in the market today. The team has used these resources to deep dive the design of the Tesla 4680 packs along with the BYD blade and Qilin packs. The team was also able to do some physical benchmarking through means of GM battery labs, external benchmarking suppliers, and A2MAC1 software provided through GM.



*Battery pack with a cell-to-pack design and prismatic cells, illustrating the option of using fewer but larger cells than typically in packs based on cylindrical cells
(Courtesy of Henkel)*

Figure 1. Example of Cell-To-Pack Battery.

Cell-to-pack battery architectures do have their fair share of challenges when it comes to ramping up their production levels. Particularly, their safety. Cell-to-pack designs can remove failure-mode mitigations, specifically around venting and thermal propagation. As for cooling, most of the cells would be laid on a cooling plate, similar to module based cells. The cooling plate transfers heat away from the cells through both conduction and convection. The plate absorbs some heat on the surface,

and also has cooling lines that circulate a coolant that dissipates the heat. However, since the cells are more densely packed, there is a higher energy density which can lead to faster and more severe thermal runaway events. Prevention of thermal propagation and mitigation of it when it does happen is critical in the development of cell-to-pack architectures. Additionally, the cell to pack battery designs all rely heavily on dispensed potting, thermal interface materials and adhesives. These materials add complexity to the manufacturing build process while introducing both design and process failure modes that need to be properly mitigated. Finally, due to the adhesives the cell to pack batteries severely limits the components that can be serviced in the event of a failure. This significantly increases the cost associated with component failures and requires a detailed failure rate analysis to weigh the initial product cost against the service and manufacturing scrap costs.

1.3 Stakeholder

The primary stakeholder for this project is the New Products Battery Architecture team. Specifically, the organization director, engineering group manager and future battery development team. They will be the customer of the report.

Secondary stakeholders include the GM organizations that support and participate in the design, development, manufacturing and product execution of the future battery and vehicle program. These stakeholders include:

- Customer - Person purchasing the vehicle
- Manufacturing - Vehicle Assembly and Battery Assembly Teams
- Product Teams - Battery and Vehicle Chief Engineers
- Battery Cell Engineering
- Hardware - Battery Component Execution
- Battery Sensing and Controls
- Software - Embedded Controls
- Battery Component and Vehicle Systems Safety
- Vehicle Integration
- Battery and Vehicle Service
- Battery Recycling
- Product Quality

1.4 Scope

The scope of this project is to create a module based battery pack design using NX and assess the design against the currently developed cell to pack design. The generated design will be assessed against the same requirements and specifications of the current cell to pack design. To manage the project timeline the detail of the design will be simplified and the analysis will use synthesis vs detailed CAE models. Surrogate data, GM technical specialist guidance and benchmarking will be leveraged where possible.

1.5 Project Deliverables

Create a module based battery pack design including the following subsystems: Module, Thermal, Electrical, Structure and Thermal Runaway Protection. For the design, summarize the following criteria as compared to the cell to pack design:

1. Piece cost (non-cell)
2. GM battery pack and module manufacturing cost implications
3. Pack mass
4. Pack energy
5. Service cost
6. Recycling Cost

Deliverables 1, 2, 5 and 6 provide the enterprise cost or “life cycle assessment” for the battery pack architecture. This life cycle assessment is depicted below. Mining and processing of raw materials is out of scope for the project. The cost associated with getting the raw materials into the final product state of the parts is built into the piece price of the parts supplied to GM.

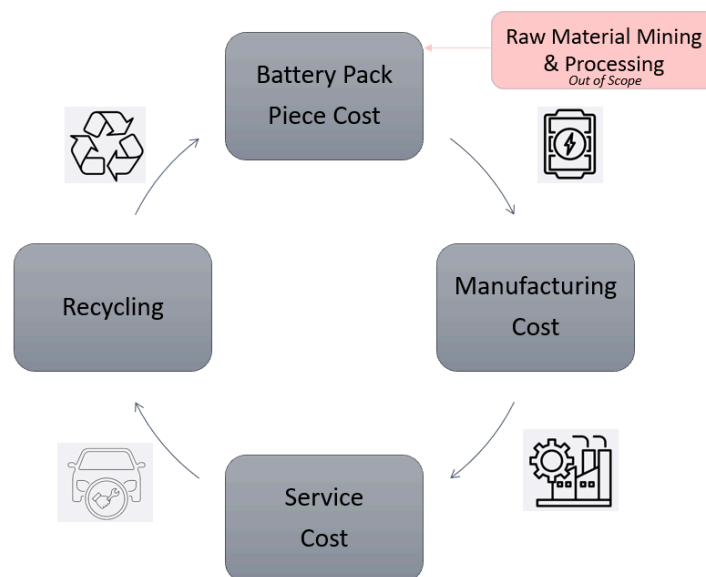


Figure 2. Lifecycle of EV battery.

2.0 Assumptions and Methodology

The approach for each deliverable is highlighted below. These approaches will not be of the same depth as a standard GM project due to team size and resources but will closely follow the same standard process the new products battery architecture team executes.

1) For the engineering cost estimates and standard GM finance piece cost estimation The team will distribute the components among ourselves. One third of the parts to each person and work with cost engineers at GM for estimates. The team will also heavily rely on prior component cost estimates and supplier quotes for similar part designs.

2) For the manufacturing investment and floor space estimation the team will distribute the ME floor space and investment assessments. Each person will perform at least one. Quantity of assessments will be determined by the design during the project. Prior battery program plant layouts and cost estimates will be utilized to capture direction cost savings or risks for each design. Additionally, prior program trade off studies for Manufacturing implications of cell to pack will be utilized.

3, 4, 7) Team will perform trade off studies via Pugh format to select designs. Once the designs are complete we will conduct virtual analysis through the GM virtual development and validation organization. The team will distribute components/subsystems between each other. Will assign subsystems during the first part of the class. Target assignments will be mid September. Subsystems: Module, Structure, Thermal, Thermal Runaway, Electrical. For thermal runaway performance prior analysis from similar designs will also be utilized for estimating expected performance.

5) CAD based mass estimations The team will distribute the components among ourselves. One third of the parts to each person. The team will create the design using NX. The design work will be instructed by the team members and executed through the support of designers in the advanced battery design organization.

6) CAD based assessment of cell packaging area using cell Wh/L will be done through simple volumetric calculations in excel and NX for complex geometries.

10, 11, 12) Excel assessment to requirement lists. The team will distribute the components among ourselves. One third of the parts to each person and use checklists to assess compliance.

8) To identify the service cost the team will rely on our internal team's service expertise, consult with GMs service organization and utilize service cost studies on prior programs. Additionally, the team will consult with GMs advanced battery organization and utilize data from an advanced technology work study that focused on the service cost implications for all battery hardware. This study provides the input incident per thousand vehicles (IPTV) data to perform design specific assessments. These are defined for 12 months in service up to 96 months in service. It is also defined for year one to year 10 in service which incorporates how IPTV reduces with continuous design improvements.

3.0 Results and Discussion

The team developed two module based battery pack architectures to compete with the cell to pack baseline. Both architectures utilize the same exterior battery enclosure envelope as the cell to pack design. The difference in the module architectures lies in the mounting strategy. Strategy one is a flange mounted design and the second mounts the module via two end plates and a mid plate. These two architectures have been executed in production by multiple EV manufacturers and EV battery suppliers. An example of the mid plate battery design is shown in Figure 3. The modules are not pictured, but the end plates and mid plates are in place to show the mounting locations.

The modules are not pictured, but the end plates and mid plates are in place to show the mounting locations. The team has designed the flange mounted architecture in NX. The actual design is considered confidential. Thus for this report, industry examples will be shown in place of the GM design. Additionally, for the UofM report exact numbers are replaced with relative metrics to the baseline cell to pack design.



Figure 3. Example of a flange mounted battery pack with long fore/aft modules
*GM Capstone Design not shown due to confidentiality.

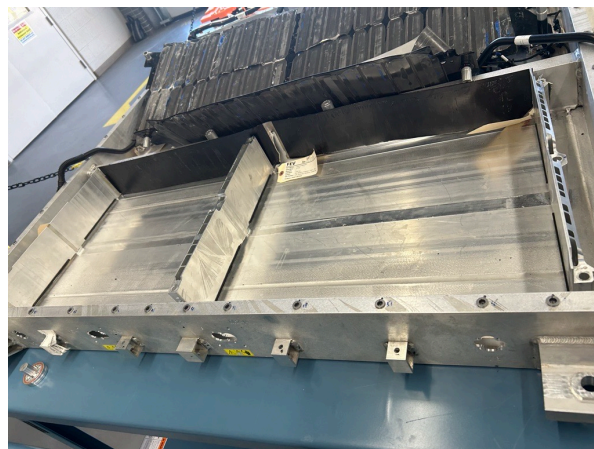


Figure 4. Example of a mid plate battery pack.

The results and discussion section is limited to the flange mounted design which has been designed in NX. The end/mid plate mounted design is directionally equivalent to that of the flange mounted design but enables a higher packaging efficiency and overall pack energy. The cost estimates are based on scaling the costs of other GM battery components. The reference components execute the same design with different dimensions and come from a variety of packs.

Table 2. Comparison criteria for the different pack architectures.

Criteria	Cell to Pack	Module to Pack Flange Mounted	Module to Pack End/Mid Plate Mounted
Energy	Baseline	Lowest	Lower
Mass	Baseline	Lower	
Piece Cost	Baseline	Higher	
Service	Baseline	Lower	
Manufacturing Cost	Baseline	Higher	
Manufacturing Scrap	Baseline	Lower	
Recycling	Baseline	Lower	
Enterprise Cost	Baseline	Lower	
Recommendation	Baseline	Lower cost but much lower energy	Lowest Cost with best module based energy

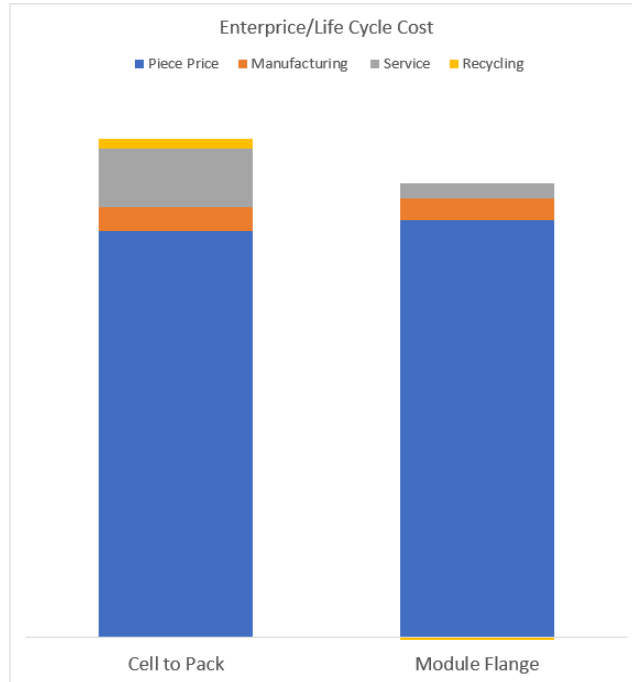


Figure 5. Enterprise/Life cycle cost assessment of cell-to-pack vs module to pack.

Table 3. Cost comparison of cell-to-pack and module architecture designs.

	Cell To Pack	Module Flange
Piece Price	+	-
Manufacturing	-	+
Service	-	+
Recycling	-	+
GM Business Case	-	+

3.1 Cost & Mass

The battery piece price is broken down into the functional subsystems: Module, electrical, structures, thermal runaway protection, thermal cooling system and cell.

Table 4. Cost of individual subsystems within the battery pack.

Pack Level Cost	Cell to Pack	Module to Pack Beam Mounted Module	Module to Pack Floor Mounted Module
Module- (Cell Block)	Baseline	+	+
Electrical		=	=
Structures		-	-
TRP		=	=
Thermal		+	+
Cell		-	-
Pack Total		+	+
Non-Cell		+	+
Delta		+	+

Table 5. Mass of individual subsystems within the battery pack.

Pack Level Mass	Cell to Pack	Module to Pack Beam Mounted Module
Module- (Cell Block)	Baseline	-
Electrical		=
Structures		+
TRP		=
Thermal		-
Cell		+
Pack Total		+

3.1.1 Structure

The first subsystem evaluated was the structure. The structure consists of three main elements. A stamped lower tub, cover and an internal structural frame. The exterior tub and cover have been kept common for all three designs to enable a direct comparison between pack architectures.

The internal structural frame for the module pack architectures has been entirely re-designed to accommodate the modules. A unique cross section for the module mounting beams was implemented to replace the cell to pack design. This cross section is a unique design developed by GM for the current Ultium products (Lyric, Blazer, Equinox, Silverado, Hummer, etc...). The beams run front to back of the battery pack surrounding three large modules. They are joined to the lower tray using L-brackets. The beams are further joined via two cross car C-channel sections which are also welded to the tray. These design changes resulted in an equivalent pack structure cost to the cell to pack design.

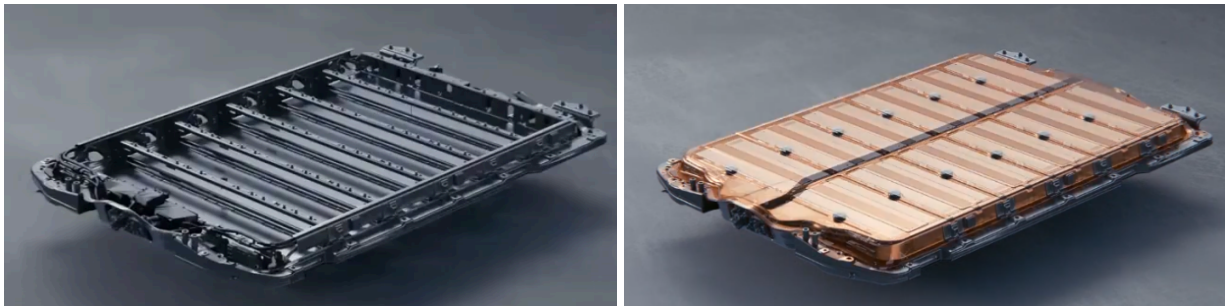


Figure 6. General Motors Ultium battery structure. Tray (Left) and cover (right)

3.1.2 Module

There are two ways to mount the modules into the pack: flange mounted or using a mid plate. The flange mounted design is aptly named as flanges from the modules attached to the beams running for/aft. The connections are spaced such that two modules side-by-side can share the same beam. The

mid plate design adds a mid plate that is used along with the end plates to mount the module. Each design will fit different size cells, and therefore carry different amounts of energy. Current estimates are that the flange design loses the most energy compared to the cell-to-pack design. The mid plate design is still a reduction from the cell-to-pack design but an overall smaller reduction. By moving forward with the mid plate design, the pack will be able to retain the most amount of energy. The comparison of energy levels is shown in Table 2.

With either the flange or mid plate design, there is a cost increase compared to the cell to pack design. Module based packs require side and end plates for each module. These parts are made of aluminum and have a substantial cost implication. This is even higher for the mid plate design due to the higher quantity of end plates.



Figure 7. Example of battery module (left) and cell array for a cell to pack design (right)

3.1.3 Thermal

In the cell to pack design there is a pack level cold plate with one inlet and one outlet. In contrast, the module based packs each have a cold plate with its own inlet and outlet. The cooling assembly for the module design also includes jumpers off the inlet and outlet lines to reach the cold plates. This adds six jumpers and each jumper adds complexity to the design as the connections may fail. This is a risk that must be mediated during the DFMEA stage of the design process in order to ensure reliability for the customer. Battery pack service costs are high due to the amount of labor hours required to remove the battery from the vehicle and replace internal components. Thus, added failure points have the potential to increase service cost substantially.



Figure 8. Example of module based cold plate from VW ID4.

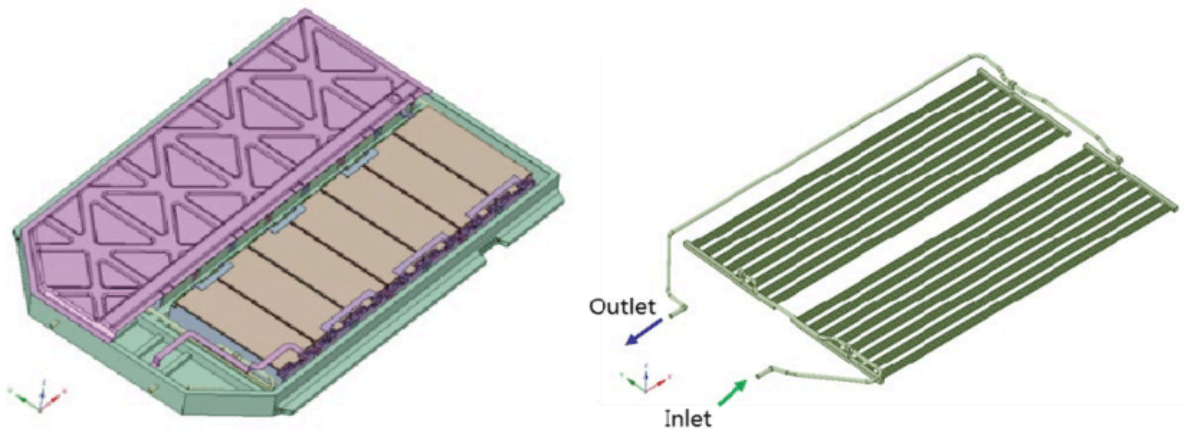


Figure 9. Example of cell-to-pack cold plate.

3.1.4 Electrical

Finally, the electrical subsystem was evaluated for the module to pack architectures. The electrical system contains the electrical busbars, battery disconnect unit (BDU), central monitoring unit (CMU) and their associated harnesses and connectors. The electrical layout is almost identical to the cell to pack baseline. Besides some minor size differences between the architectures, the main difference and cost increase comes through the module to module busbar connectors. In the module pack architecture, the busbars connect modules using a removable screw-in fastener. This is to allow for the replacement of modules in the case of a failure of one of the cells. The cell to pack architecture does not need this as the cells are assembled throughout the pack rather than in modules. This allows for the busbars connections between the structure to be connected via welds, saving cost and weight.



Figure 9. Example of a busbar used for cell-to-cell connections.

3.2 Service

One of the main differences between the cell to pack baseline and the module to pack battery comes in the serviceability of the internal cells. In the cell to pack architecture, the cells are permanently bonded to the pack, making the cells unserviceable. A failure of a single cell or any component permanently attached to the cells can ultimately result in the battery requiring to be replaced. In the cell to module, the cells are laid out in modules, which are then assembled into the pack. In the case of a single cell failure, it is possible to replace the module and not the whole pack. The improved serviceability of the cells will lead to a reduction in warranty spend utilizing GM warranty data. To make the cell to pack architecture viable, the failure rate of battery internals must be reduced.

From prior studies done at GM, going from module based to cell to pack increased the cost per vehicle. Studies were based on prior warranty data. A new estimate for our capstone design will be created but the team is confident that the module to pack will continue to provide lower service costs.

In general, servicing the RESS introduces a lot of complexity, labor and replacement cost. Starting with a cell to pack architecture, after removal from the vehicle, the failed RESS must be swapped into the shipping container with the new RESS. The cells and the ICB are deemed not serviceable in cell to pack and these components happen to have the highest failure rate. More often than not, the full pack will need to be replaced which is where a majority of the enterprise cost is introduced.

For module-to-pack cells, the cell modules are deemed serviceable so this allows for a bit more flexibility in servicing the RESS. Once the RESS is out of the vehicle, it can be opened, disassembled to allow for the replacement of the failed module. Then it is assembled and sealed back up to allow for the reuse of the RESS. This process adds labor, however, the cost of individual modules are much less than the cost of the full RESS replacement, allowing for greater enterprise savings given the failure rate seen and future failure rate that was estimated. This is summarized in table 6 below.

Table 6. Section Replacement vs. Full Pack Replacement

	Full Pack	Section Replacement
Labor Cost	Slightly Lower	Slightly Higher
Pack Part Cost	Much Higher	Much Lower
Total Cost Per Vehicle	Higher	Lower

Updated total RESS IPTV data was gathered and will be used in the updated service study. As the baseline, it is assumed that the entire battery is replaced in the instance of a failure. The total RESS IPTV data is shown below in Table 7; green indicating low IPTV, red indicating high IPTV. Using the IPTV data by component, the team was able to average out the cell and ICB IPTV and create a percentage of the vehicles affected by the cell and ICB failure.

Table 7. Total RESS IPTV Data; Green = low, Red = high

RESS Total IPTV								
Year	12	24	36	48	60	72	84	96
1								
2								
3								
4								
5								
6								
7								

Next, for the module to pack architecture, the modules and the ICBs are deemed serviceable as mentioned above. The IPTV data was then gathered specifically for the modules and ICB failures. The amount of module and ICB failures projected over the total volume of the vehicle shows that a majority of the packs will fail and require a module or ICB to be replaced. The IPTV of the module and the ICB represents battery packs that could be serviced if in a module to pack architecture. The updated IPTV for the module and ICB can be seen below in table 8. Looking at table 8, the same IPTV trend is seen showing that a large majority of RESS failures can be serviced if the cells are serviceable. This leads to a much smaller cost for service in the fact that the company would not have to replace entire packs, instead they can replace the module within the failed pack. **The module based will save GM a significant amount of cost given that ICBS and modules are much cheaper than the total pack cost.**

Table 8. Module and RESS IPTV; Green = low, Red = high

IPTV from Cell & ICB Assembly Total								
Year	12	24	36	48	60	72	84	96
1								
2								
3								
4								
5								
6								
7								

For the module to pack architectures that the team has designed, there was a concern in the size of the modules and how a technician would be able to remove them. Service enablers would be a midplate for cutting the modules in half, or creating a lifting feature to aid technicians in removal.

There are a lot of other intangible items that if acted on can enable this serviceability of battery packs at dealerships including proper technician training, personal protective equipment (PPE) and battery design to mitigate arcing and safety risks seen by the technicians. Currently, GM technicians work on the Bolt EV batteries but not the current production Ultium packs. The cost savings can only be realized if technicians are enabled to perform the service procedures with the proper safety precautions and design for safety and serviceability.

Finally, there are a lot of uncertainties at the dealership and customer level that should be considered. If GM goes with the cell to pack design, post-warranty battery strategies should be considered to ensure that customers are kept for life. In talking with experts close to service, some ideas would allow the customer to get a new battery at a reasonable price or have GM buy their vehicle back to recycle and give the customer a deal on a new EV. Another talking point is GM's strategy to ensure that the battery controller and other high IPTV serviceable components are considered serviceable. In states where repeat warranty claims/recalls happen, if the controller is serviceable, that will be high and recalled, rather than the whole battery pack. If the cell to pack makes the controller or other internal components not serviceable this could lead to GM recalling a lot of batteries.

3.3 Manufacturing

The manufacturing assessment includes the floor space cost, equipment investment cost, and the scrap cost. Based on two prior internal assessments on both SUV and truck packs, the scrap costs of cell to pack are higher than that of the module based packs while the floor space and investment are lower.

The manufacturing floor space and investment savings are a result in reduced part count and the associated assembly steps. Specifically, the removal of module structure (side plates, end plates and cover) and cooling parts (module cold plates, inlet and outlet connections).

The scrap costs for a cell to pack architecture are much higher. This is a result of the cell to pack design bonding the cells directly into the pack structure. Once the bond to the structure is made these parts are no longer repairable and nearly a full pack can be required to be scrapped. This is in contrast to a module design where the scrap can be limited to a smaller group of cells (module). For this study both the cell to pack and the module designs consist of three cell groupings. Thus the scrap costs for a defect in one cell group are one third the cost of the cell to pack.

3.4 Recycling

When evaluating the recycling of the battery pack there are many factors, including logistics and processing, but the dominant factor when comparing cell to pack vs module to pack is the disassembly cost. The current disassembly processes are manual and thus the number of hours to disassemble the pack into shreddable pieces impacts the potential cost or value in recycling.

A conventional module based battery pack design can be disassembled relatively quickly. The conventional design here refers to a group of cells in a module that is bolted into the pack structure. The pack structure consists of a tray and cover which can either be disassembled by cutting an dispensed style seal or removing fasteners for a press in place style seal.

In contrast, a conventional cell to pack design takes considerably more time to disassemble. A conventional cell to pack design here refers to a group of cells that are bonded together and to the lower tray and/or cover of the pack structure. The bond strength between the cells and the pack structure can range considerably. A PET film bond on a cell can be as low as a fraction of an MPa (values removed for UofM report). A thermal interface material can range in the low MPa values, while an adhesive can have high MPa values

From personal experience disassembling the Tesla Model Y pack, discussions with A2Mac1 and Caresoft for Tesla, BYD and CATL designs the processes of opening a cell to pack and removing cells is considerably longer than a module based pack. A conservative estimate could be greater than a full day with multiple operators to an aggressive estimate of slightly less than one full work day.

It is important to note that as the market matures the recycling suppliers will adapt their processes and equipment to reduce the disassembly time and increase the size of the shredding machines. From

discussions with recycling suppliers at the Detroit Battery they are working on technologies and processes for addressing the cell to pack designs.

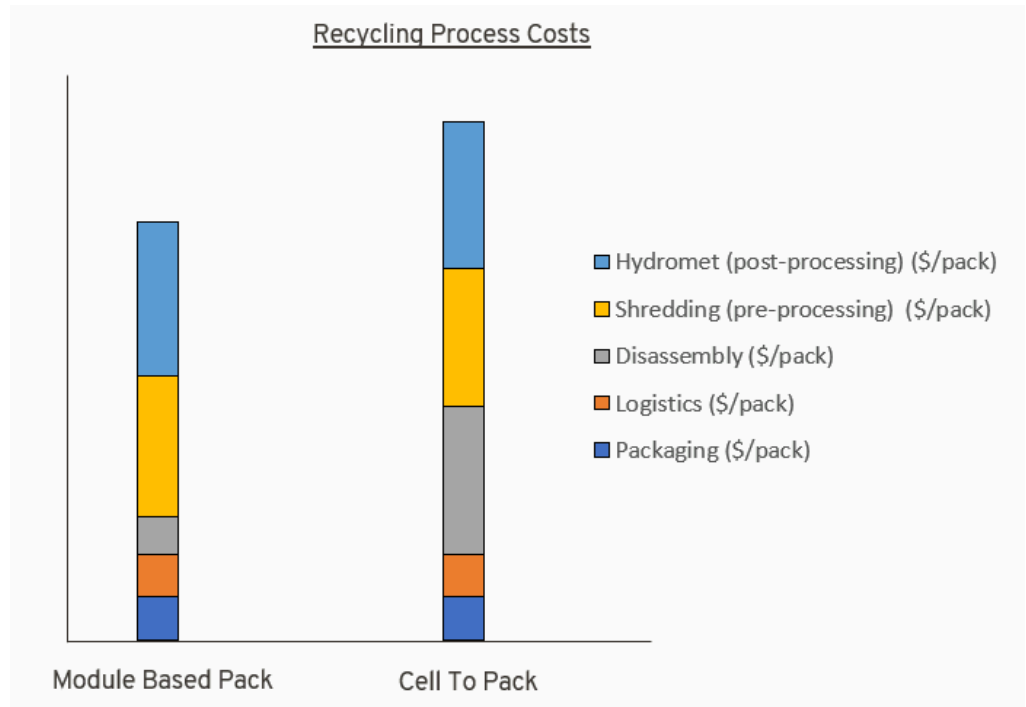


Figure 10. Recycling value compared to piece cost of module and cell to pack batteries.

4.0 Recommendations

The team evaluated the lowest enterprise cost approach for designing, producing, servicing and recycling an electric vehicle battery pack for high volume production. This assessment included key specifications of the battery pack and the associated costs over the life cycle of the product.

4.1 Battery Specifications and Piece Price

Integrating the cells directly into the structure of the battery pack eliminated the need for separate modules which reduced the number of components and the design complexity. The volumetric space savings resulted in the highest energy solution and lowest piece price.

The cell to pack design is particularly beneficial as it removes the need for module structure. Contrary to the initial expectation the cell to pack, pack level, cold plate resulted in a higher cost and mass solution than the module based pack. This is a result of the pack level cold plate covering a larger area of the battery pack.

Overall, the lower piece price solution is the cell to pack architecture. The cell to pack provides more energy but can result in a similar or higher mass than the module based architecture.

4.2 Battery Manufacturing

The manufacturing assessment covered two topics: the cost of the plant and the scrap cost associated with building the battery. The manufacturing line for a cell to pack architecture is more efficient in terms of material usage and assembly time as it reduces the number of assembly stations (steps) required to assemble the battery. The fewer number of components lowers the amount of labor or automation required. However, assembling the cells directly into the structure requires specialized equipment and processes that can be capital and development intensive.

The module to pack approach, while it involves additional assembly stations, relies on an existing well known set of processes and equipment. Additionally, the cost implications of assembly defects is significantly reduced. A defect in the module assembly resulted in scrapping, at worst, one module or a third of the cells in a pack. The cell to pack design, however, could result in scrapping nearly an entire battery pack with each defect.

Overall, from a manufacturing cost perspective, the module based architecture provides the lowest cost.

4.3 Battery Service

The largest cost implication of all the life cycle steps comes from the service implications. In the event of an in-vehicle-life failure, servicing a cell to pack system is more complex and costly. The cells being integrated directly into the tray means they cannot be individually replaced or serviced without disassembling the entire pack. This drove significantly higher labor time and replacement costs. Even when only considering the IPTV of the hardware permanently attached to the cells. The module to pack design reduced both the labor time and replacement costs due to the ability to swap out one section of a battery pack instead of the entire pack.

However, it is important to note that to achieve the full benefits of a module based battery pack, from a service perspective, the industry needs properly trained technicians and the battery pack needs to be designed for safe disassembly and maintenance.

Overall, the module based battery pack provides the lower cost solution compared to the cell to pack design.

4.3 Battery Recycling

Similarly to service, the cell to pack design suffered from high disassembly labor costs which also resulted in the module based design providing the lowest cost solution.

With this conclusion, it is still important to highlight that the cell to pack design does provide higher energy per pack, which can increase the value recovered per pack. Additionally, the logistics cost plays

a large role in the end of life cost and the pickup/delivery of battery packs will need to be coordinated to minimize those expenses.

Overall, the module based design provides the lowest cost recycling solution with the current state of the industry.

4.4 Recommendation & Conclusion

While the industry trend is moving towards more cell to pack batteries to achieve improved up front piece cost and battery specifications the manufacturing scrap, service, and recycling costs are key risks to that integration strategy.

Unless manufacturing defects, in-service IPTV, and lower cost recycling technologies are developed, the lowest enterprise cost solution today is the module based battery pack.

5.0 Impact/Financial Benefits

The overall financial benefit for executing the module based battery pack design, when including all the life cycle costs, divided over the volume of the program is \$142.3M/year (or \$952/pack).

6.0 Summary/Conclusions

The battery architecture plays a dominant role in electric vehicle profitability. The two leading battery architectures in the market today are the cell to pack and the module based designs. Each architecture offers distinct advantages and disadvantages in terms of up front cost, manufacturing, service, and recycling.

The cell to pack design has recently become recognized for being a lower piece cost solution due to the reduced number of components. This benefit has started a trend towards this architecture which is in production today from multiple manufacturers including: Tesla, BYD, and CATL. The reduced number of parts translates directly into fewer manufacturing steps, which in turn reduces the number of stations needed in the production process. With fewer stations the manufacturer can optimize the production process, reduce floor space, and the required capital investment in equipment.

Additionally, the reduced structure around the cells results in the ability to package more capacity. This battery capacity will provide improved charge and discharge performance along with increased vehicle range. Therefore, depending on the product requirements, this benefit may or may not be needed to meet the product capacity and performance requirements.

However, the cell to pack design also has key disadvantages in terms of manufacturing scrap, service, and recycling. The bonding of the cells directly to the battery structure either increases the difficulty or entirely removes the ability to replace a set of cells. As a result, either the entire battery or the majority

of a battery pack may be required to be replaced if there is a failure in one of the parts that is permanently attached to the cells. This trade off impacts the cost of manufacturing scrap during initial production, the cost of warranty repairs, and the cost of recycling.

Therefore, while the cell to pack design offers lower upfront costs, the module based design is more advantageous for those looking to minimize the lifecycle costs of the product, assuming the module based design can meet the performance and capacity requirements.

In conclusion, given the current state of manufacturing defect rates and in-service IPTV projections the module based design is recommended to enable sustainable EV manufacturing by minimizing the lifecycle costs. A cell to pack design may however be an attractive alternative if necessary to meet product performance requirements.

7.0 Appendix

Tables A1 and A2 below are the values utilized in the energy calculations for the battery architectures.

Table A1. Dimensions of current BEV4 cell-to-pack design.

Dimension	Value
Cross-Car Beam Thickness [mm]	Value 1
Nominal Gap from Cell to Beam [mm]	Value 2
SAM Thickness [mm]	Value 3
Side Plate Thickness [mm]	Value 4
Cold Plate Gap [mm]	Value 5

Table A2. Energy calculation for mid-plate design, flange, and comparison to current cell-to-pack.

	Mid-Plate Design	Flange Design	Cell-to-Pack Design
Beams Total Cross-Car Space [mm]	Value 1	Value 12	Value 23
Cross Car Gap Space [mm]	Value 2	Value 13	Value 24
Side Plate Thickness [mm]	Value 3	Value 14	Value 25
SAM Gap [mm]	Value 4	Value 15	Value 26
Total Non-Cell Width Space [mm]	Value 5	Value 16	Value 27
Cell Space [mm]	Value 6	Value 17	Value 28
Cell Width [mm]	Value 7	Value 18	Value 29
Cell Width (Rounded for NX) [mm]	Value 8	Value 19	Value 30
Cell Height [mm]	Value 9	Value 20	Value 31
Pack Energy [kWh]	Value 10	Value 21	Value 32
Cell Energy [kWh]	Value 11	Value 22	Value 33

Service and recycling estimates are based on the incident per thousand vehicles (IPTV). These are defined for 12 months in service up to 96 months in service. Two sets of data were used. The first set evaluates the IPTV on a component basis in detail. The second provides a pack level IPTV only.

Table A3.

Component	Latest Program Assessment (BET)		
	12 MIS	24 MIS	36 MIS
CMU Hardware	Value 1	Value 22	Value 43
LV Harness (Module Com)	Value 2	Value 23	Value 44
External Short blow BDU fuse	Value 3	Value 24	Value 45
Cell	Value 4	Value 25	Value 46
ICB	Value 5	Value 26	Value 47
BRFM	Value 6	Value 27	Value 48
Bus Bar	Value 7	Value 28	Value 49
Software	Value 8	Value 29	Value 50
BDU	Value 9	Value 30	Value 51
Cooling System	Value 10	Value 31	Value 52
FPC	Value 11	Value 32	Value 53
TRP Sensors	Value 12	Value 33	Value 54
HV Connectors	Value 13	Value 34	Value 55
External	Value 14	Value 35	Value 56
Tray/Cover/Seals/Structure	Value 15	Value 36	Value 57
MFG/Handling	Value 16	Value 37	Value 58
MISC/Isolation Fault	Value 17	Value 38	Value 59
CMA (-CMU/ICB/cell)	Value 18	Value 39	Value 60
Hardware Total	Value 19	Value 40	Value 61
Cell & ICB	Value 20	Value 41	Value 62
% of affected vehicles	Value 21	Value 42	Value 63
Cell and ICB% Average	Average		

Table A4.

RESS Total IPTV								
Year	12	24	36	48	60	72	84	96
1	Value 1	Value 8	Value 15	Value 22	Value 29	Value 36	Value 43	Value 50
2	Value 2	Value 9	Value 16	Value 23	Value 30	Value 37	Value 44	Value 51
3	Value 3	Value 10	Value 17	Value 24	Value 31	Value 38	Value 45	Value 52
4	Value 4	Value 11	Value 18	Value 25	Value 32	Value 39	Value 46	Value 53
5	Value 5	Value 12	Value 19	Value 26	Value 33	Value 40	Value 47	Value 54
6	Value 6	Value 13	Value 20	Value 27	Value 34	Value 41	Value 48	Value 55
7	Value 7	Value 14	Value 21	Value 28	Value 35	Value 42	Value 49	Value 56

Table A5.

IPTV from Cell & ICB Assembly Total										
Volume	Year	12	24	36	48	60	72	84	96	Total
Year 1 Volume	1	Value 1	Value 8	Value 15	Value 22	Value 29	Value 36	Value 43	Value 50	Volume * Value 50
Year 2 Volume	2	Value 2	Value 9	Value 16	Value 23	Value 30	Value 37	Value 44	Value 51	Volume * Value 51
Year 3 Volume	3	Value 3	Value 10	Value 17	Value 24	Value 31	Value 38	Value 45	Value 52	Volume * Value 52
Year 4 Volume	4	Value 4	Value 11	Value 18	Value 25	Value 32	Value 39	Value 46	Value 53	Volume * Value 53
Year 5 Volume	5	Value 5	Value 12	Value 19	Value 26	Value 33	Value 40	Value 47	Value 54	Volume * Value 54
Year 6 Volume	6	Value 6	Value 13	Value 20	Value 27	Value 34	Value 41	Value 48	Value 55	Volume * Value 55
Year 7 Volume	7	Value 7	Value 14	Value 21	Value 28	Value 35	Value 42	Value 49	Value 56	Volume * Value 56
										Lifecycle Total
										Sum of totals

8.0 References

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